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Detection of Bangla AI-Generated Texts Using a Quantum Kernel Support Vector Machine

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ABSTRACT

Machine-generated text is increasingly hard to distinguish from human writing, raising concerns for academic integrity, journalism and information trust, especially for low-resource languages such as Bangla, where detection tooling lags far behind English. This paper presents, to the best of our knowledge, the first application of a quantum-kernel Support Vector Machine (QSVM) to detecting AI-generated Bangla text. On a class-balanced Bangla dataset spanning six genres (newspaper articles, scientific abstracts, emails, poems, short stories and office orders) that is strongly imbalanced across genres, we extract BanglaBERT contextual embeddings, reduce them to eight principal components and map each sample onto an 8-qubit Hilbert space with an angle-embedding feature map and a ring of entangling CZ gates. The fidelity between mapped states forms a quantum kernel that is passed to a classical SVM. On a held-out test split, the QSVM attains 95.0% accuracy, 95.0% F1-score and 0.989 ROC-AUC. Under the same 5-fold stratified cross-validation protocol applied to all models, it achieves 95.2% (±2.8%) mean accuracy, compared with 95.0% (±2.0%) for Logistic Regression and 93.8% (±2.4%) for an RBF-kernel SVM on identical features and 85.8% (±4.1%) for a Variational Quantum Classifier (VQC). Replacing BanglaBERT with TF-IDF features lowers the QSVM's cross-validated accuracy to 93.7% (±2.7%). Within the specific configurations evaluated, the QSVM was more accurate and less variable than the single VQC architecture tested; the experiments do not identify the cause of this gap. Per-genre results are reported alongside the aggregate metrics. These results provide a first quantitative benchmark for quantum-enhanced Bangla AI-text detection and indicate that the QSVM is competitive with, rather than superior to, strong classical baselines.

1 | Introduction

The rapid advancement of large language models (LLMs) such as GPT-3.5/4, LLaMA and their successors has made machine-generated text virtually indistinguishable from human writing in many everyday contexts. While this capability unlocks substantial productivity gains, it also introduces serious risks such as academic dishonesty through AI-written assignments, erosion of trust in journalism and online reviews and large-scale misinformation campaigns. Consequently, the automatic detection of AI-generated text has become an active and fast-moving research area, with detectors ranging from zero-shot statistical methods [1–3] to supervised transformer classifiers [4–6] and adversarially robust frameworks [7, 8].

The overwhelming majority of this literature, however, targets English or other high-resource languages. Bangla (Bengali) is spoken natively by over 230 million people, making it one of the most widely spoken languages in the world, yet AI-text detection research for Bangla remains sparse. The studies closest to this gap, BanglaTextDistinguish [9] and a subsequent comparative study of zero-shot versus fine-tuned transformers on

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the same corpus [10], rely on a single ChatGPT-paraphrased corpus and purely classical architectures. A separate line of work introduced a genre-diverse, class-balanced Bangla AI-text dataset spanning newspaper articles, scientific abstracts, emails, poems, short stories and office orders [11], which is the dataset used in this study.

In parallel, quantum machine learning (QML) has emerged as a candidate paradigm for classification tasks, motivated by the ability of quantum feature maps to embed classical data into exponentially large Hilbert spaces that may be intractable to reproduce classically [12, 13]. Early applications to natural language processing such as sentiment classification [14], short-text classification [15] and quantum kernel design for compositional semantics [16], report encouraging, if still small-scale, results and recent surveys note that QNLP remains constrained chiefly by limited datasets and near-term quantum hardware rather than by a lack of theoretical promise [17]. To date, however, no prior work has applied quantum machine learning, variational or kernel-based, to Bangla AI-generated text detection specifically. Our research addresses that gap.

Our contributions are as follows:

1. We present the first application of a quantum kernel Support Vector Machine (QSVM) to Bangla AI-generated text detection, using a fixed 8-qubit angle-embedding feature map with entangling CZ gates and a fidelity-based kernel evaluated on PennyLane's `default.qubit` simulator.
2. We design a hybrid classical–quantum pipeline that couples BanglaBERT contextual embeddings with PCA-based dimensionality reduction, making dense, semantically rich features tractable for a small-qubit-count simulator.
3. We provide a comparison against a Variational Quantum Classifier (VQC) trained on the identical feature pipeline and against classical Logistic Regression and RBF-SVM baselines, all evaluated on the same features and under the same 5-fold protocol so that differences reflect the classifier or kernel rather than preprocessing.
4. We report 5-fold stratified cross-validation results for every model in addition to a single held-out split, a BanglaBERT-versus-TF-IDF ablation for both the QSVM and the VQC and per-genre results and release the full implementation for reproducibility.

2 | Related Work

2.1 | AI-Generated Text Detection

Early AI-text detectors relied on zero-shot statistical signals derived from a target language model's own probability distribution. DetectGPT [2] exploits negative curvature in a model's log-probability function under small perturbations, while DetectLLM [3] uses log-rank statistics to achieve comparable accuracy with far fewer perturbations. Cross-domain robustness has also been studied directly. Rodriguez et al. [1] show that a RoBERTa-based detector can be adapted across scientific domains with only 200–500 in-domain labels, though performance degrades sharply on short (< 500-character) passages, a limitation echoed across nearly every detector surveyed here [5, 8, 18].

Supervised and ensemble approaches remain the dominant paradigm for in-domain performance. Ghostbuster [6] combines per-token probabilities from multiple reference language models with a logistic regression classifier to reach 99.0 F1 in-domain, while MGTBench [5] and M4 [18] benchmark a wide range of metric-based and model-based detectors and consistently find that in-domain accuracy near 99% collapses under distribution shift, to unseen generators, unseen domains, or adversarial paraphrasing. RADAR [7] directly targets this weakness with an adversarially trained detector–paraphraser pair, improving AUROC by up to 31.6 points against unseen paraphrasers and OUTFOX [19] uses an attacker–detector in-context-learning loop to similarly harden detection without parameter updates. Most recently, DetectRouter [20] reframes zero-shot detection as a prototype-based routing problem to mitigate surrogate–source misalignment and SOGPTSpotter [21] applies a Siamese triplet-loss architecture to detect ChatGPT-generated Stack Overflow answers with 97.7% accuracy. A persistent theme across this literature [22–24] is that detectors trained on static datasets generalise poorly to newer, unseen generators, motivating continual benchmark renewal, exactly the concern that dataset-introduction work such as [11, 25] addresses for Bangla and English, respectively.

2.2 | AI-Generated Text Detection in Bangla

Bangla-specific detection research is considerably less mature. Raya et al. [9] introduce BanglaTextDistinguish, a 6644-instance corpus of human-written and GPT-3.5-paraphrased Bangla text and propose a hybrid BiLSTM–SVM classifier reaching 82.83% accuracy. Building on the same corpus, Islam et al. [10] benchmark zero-shot and fine-tuned transformers such as XLM-RoBERTa-Large, mDeBERTaV3, BanglaBERT, IndicBERT and Multilingual BERT, finding that zero-shot inference is near chance-level (≈50% accuracy) while fine-tuning lifts XLM-RoBERTa-Large to 91.5% accuracy. Both studies, however, are limited to a single generation source (ChatGPT paraphrasing) and a narrow genre mix. The dataset used in the present study [11] instead spans six genres such as newspaper articles, scientific abstracts, emails, poems, short stories and office orders, with balanced human-written and AI-generated classes, offering broader genre coverage for evaluating detector generalisation within Bangla.

2.3 | Bangla NLP Infrastructure

The feature-extraction backbone used in this work, BanglaBERT [26], is an ELECTRA-style discriminator pretrained with a Replaced Token Detection objective on 27.5 GB of crawled Bangla text ("Bangla2B+") and reports state-of-the-art results across the Bangla Language Understanding Benchmark (BLUB) covering sentiment classification, NER, NLI and question answering. Its embeddings have since been adopted as a standard building block

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across Bangla NLP: transformer ensembles for cyberbullying detection [27], graph-augmented BertGCN architectures for text classification [28] and hybrid stacking classifiers for large-scale news categorisation [29] all report BanglaBERT-derived representations as competitive or superior to purely statistical (TF-IDF) features. This established performance motivates its use as the semantic feature extractor in our pipeline, ahead of the quantum feature map.

2.4 | Quantum Machine Learning for NLP

Quantum kernel methods originate from Havlíček et al. [12], who show that a quantum feature map can embed classical data into a Hilbert space via a parameterised circuit, with the kernel between two samples given by the fidelity (state overlap) of their embedded states, that is a formulation later generalised by Schuld and Killoran [13] as "quantum machine learning in feature Hilbert spaces." Because this kernel requires no trainable circuit weights, no variational optimisation loop is needed, which removes the trainability problems—barren plateaus and poor local minima [30]—that can affect Variational Quantum Classifiers (VQCs) trained by gradient descent on circuit parameters. Fixed kernels have scaling difficulties of their own, however: for expressive feature maps the kernel values can concentrate exponentially with the number of qubits [31]. Whether either effect matters at the 8-qubit scale used here is not tested in this paper.

Applications to language tasks remain limited in scale. Wright [16] designs a quantum kernel from the DisCoCat compositional-semantics model for a small meaning-classification task; Alexander and Widdows [15] combine an amplitude-encoded feature map with a QSVM to reach 62% accuracy on 50 real movie reviews, noting this already exceeds prior QNLP results at that scale. Ruskanda et al. [14] report 93.3% accuracy with a Quantum-Enhanced SVM for sentiment classification using SPSA-optimised circuits and outside of text, quantum kernel SVMs have shown competitive or superior performance to classical baselines on small, high-dimensional datasets in speech [32] and other domains, attributed to the quantum feature map's ability to extract discriminative structure from limited samples. A recent comprehensive review of QNLP [17] confirms that quantum kernel methods (QSVM) are a recognised, actively studied category within QNLP, but notes that empirical results on real, non-toy text classification datasets and particularly on low-resource languages remain scarce. Closest to the present setting, Ansary [33] applies a 4-qubit trainable parameterised quantum circuit (PQC), a variational, rather than kernel-based, formulation, to detect AI-generated English scientific abstracts, reporting 85.4% accuracy and an F1-score of 0.864 against classical and transformer baselines reaching up to 97.6% (fine-tuned BERT/RoBERTa); notably, that model achieves perfect recall (1.000) but only 0.760 precision, indicating a decision boundary skewed towards over-predicting the AI-generated class. In contrast, the VQC evaluated in the present paper makes nearly equal numbers of false positives and false negatives (7 and 8; Figure 2), so that pattern is not reproduced here. This paper contributes a complementary result for Bangla, using a fixed fidelity-based quantum kernel rather than a trainable variational circuit.

TABLE 1 | Genre distribution of the dataset (each genre is split 50/50 between HWT and AGT).

Genre	Texts	% of corpus
Newspaper article	318	53.0%
Scientific abstract	162	27.0%
Email	54	9.0%
Poem	30	5.0%
Short story	30	5.0%
Office order	6	1.0%
Total	**600**	**100.0%**

Note: The bold text represents the overall dataset statistics.

3 | Dataset

This study uses the Bangla AI-generated text dataset introduced in [11], available at a public GitHub repository. Each record is labelled as either Human-Written Text (HWT) or AI-Generated Text (AGT) and the dataset is exactly class-balanced (50.0% HWT/50.0% AGT). All experiments and comparisons reported in this paper, including the baseline comparison in Section 6.5, are conducted on this publicly available corpus. Unlike prior Bangla AI-text corpora that draw from a single genre or a single generation method [9, 10], this dataset spans six distinct genres—newspaper articles, scientific abstracts, emails, poems, short stories and office orders—with each genre balanced equally between the two classes (Table 1), which broadens the range of registers covered relative to single-genre corpora; the strong imbalance in the number of texts per genre, discussed below, limits what can be concluded about robustness across genres.

The dataset is strongly imbalanced across genres: newspaper articles (318 texts) and scientific abstracts (162 texts) together make up 80% of the corpus, poems and short stories have 30 texts each and office orders only 6 (three HWT and three AGT). Aggregate metrics are therefore dominated by the two largest genres and any statement about behaviour across genres has to be read against the per-genre results and confidence intervals of Section 6.4, which show how little the smallest genres can support.

Text lengths vary considerably across the corpus, from 72 to 8184 characters (mean 805, median 579 characters), reflecting the natural variation across genres—office orders and short emails are brief, while newspaper articles and scientific abstracts run substantially longer. No records contained missing text or label fields, so no rows required removal during preprocessing.

4 | Proposed Methodology

Here, raw Bangla text is cleaned and normalised, encoded into a dense semantic representation via BanglaBERT, compressed to a small number of principal components matching the qubit budget, mapped into an 8-qubit Hilbert space by a fixed quantum feature map and classified by a classical SVM operating

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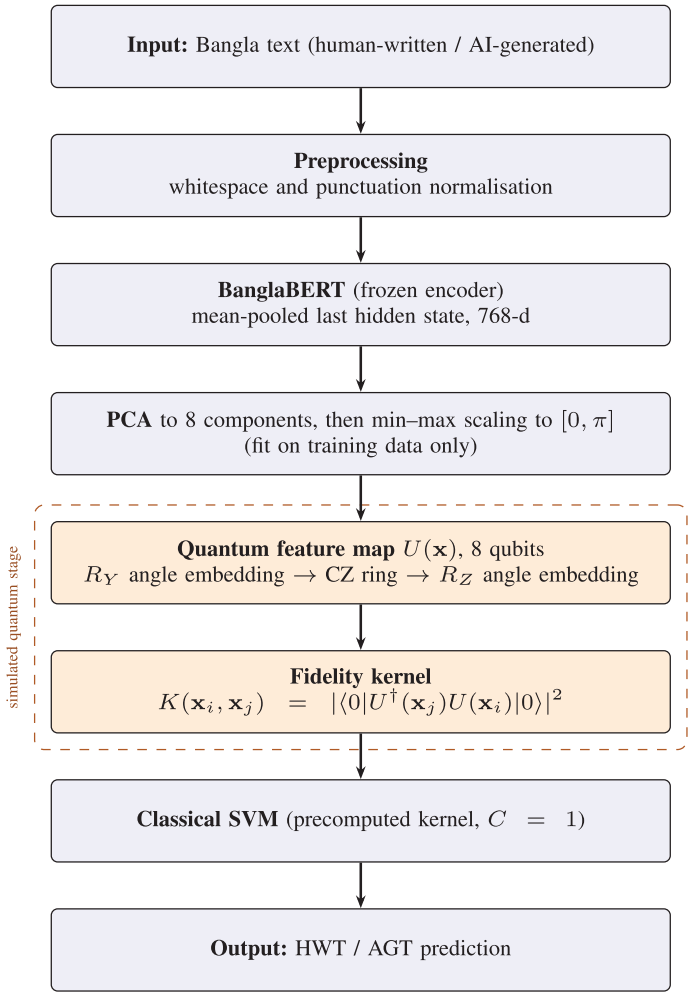


FIGURE 1 | Workflow of the proposed method: BanglaBERT → PCA → quantum feature map → fidelity kernel → QSVM. Orange blocks (dashed frame) are evaluated on a quantum-circuit simulator and contain no trainable parameters; the remaining stages are classical.

on the resulting fidelity kernel (Figure 1). Each stage is detailed below.

4.1 | Preprocessing

Each text sample is lightly normalised as carriage returns and line breaks are collapsed to single spaces, Bangla sentence-final and clause punctuation is stripped and repeated whitespace is collapsed, while the Bangla Unicode text itself, digits and basic Latin characters are preserved unmodified. This mirrors standard practice for BanglaBERT-family models, which are comparatively robust to punctuation variation but sensitive to malformed whitespace.

4.2 | BanglaBERT Feature Extraction

Each cleaned text is tokenised with the official BanglaBERT WordPiece tokenizer (vocabulary size 32,000) and passed through the pretrained BanglaBERT-base encoder [26] (an ELECTRA discriminator, 12 transformer layers, hidden size 768, 12 attention heads, sequences truncated to 128 tokens). Rather than using only the [CLS] token, we mean-pool the last hidden state across all non-padding token positions (attention-mask–weighted average), which empirically yields a more stable sentence-level representation for downstream dimensionality reduction. This produces a 768-dimensional embedding per sample.

4.3 | Dimensionality Reduction and Scaling

Simulating an n-qubit quantum circuit requires state-vector storage that scales as 2^n, which makes embedding a full 768-dimensional BanglaBERT vector directly infeasible on a classical simulator at any practical qubit count. We therefore apply Principal Component Analysis (PCA), fit on the training split only, to reduce each embedding to $n_q = 8$ components—one per qubit. On the full corpus, eight components retain 74.4% of the variance in the BanglaBERT embedding space, compared with only 13.0% when the same reduction is applied to a sparse TF-IDF representation (the 3000 most frequent word 1–2-g plus the 3000 most frequent character 3–5-g, fit on the training data only) used as an ablation baseline. Retained variance measures how faithfully the 8-dimensional projection reproduces the original representation; it is not a direct measure of class-discriminative information, so we treat the

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difference as descriptive and assess its effect on classification empirically in Section 6.3. The resulting 8-dimensional vectors are then min-max scaled to the range $[0, \pi]$ (fit on the training split, clipped on the test split) so that each component can be used directly as a rotation angle in the quantum feature map.

4.4 | Quantum Feature Map and Kernel

We define a fixed, non-trainable feature map $U(\mathbf{x})$ applied to $n_q = 8$ qubits, initialised in $|0\rangle^{\otimes 8}$:

$$U(\mathbf{x}) = \mathrm{AE}_Z(\mathbf{x}) \cdot \left[\prod_{i=0}^{7} \mathrm{CZ}(i, (i+1) \bmod 8) \right] \cdot \mathrm{AE}_Y(\mathbf{x}) \tag{1}$$

where AE_Y and AE_Z denote angle embedding of the 8 scaled features as single-qubit R_Y and R_Z rotations respectively. That is, each of the 8 scaled features is first encoded as a Pauli-Y rotation angle on its corresponding qubit, a ring of controlled-Z (CZ) gates entangles all eight qubits and the same 8 features are then encoded a second time as Pauli-Z rotation angles, following the standard construction used in quantum-kernel classification [12, 13]. The Y-entangle-Z structure ensures the feature map is non-trivial under the entangling layer (a pure Z-rotation feature map alone would commute with the CZ gates and add no expressive power).

Given two samples $\mathbf{x}_1$ and $\mathbf{x}_2$, the quantum kernel is defined as the fidelity between their embedded states:

$$K(\mathbf{x}_1, \mathbf{x}_2) = \left| \langle 0|^{\otimes 8} U^\dagger(\mathbf{x}_2)\, U(\mathbf{x}_1) |0\rangle^{\otimes 8} \right|^2 \tag{2}$$

which is evaluated by applying $U(\mathbf{x}_1)$ followed by the adjoint circuit $U^\dagger(\mathbf{x}_2)$ and measuring the probability of returning to the all-zero basis state via a Pauli-Z projector, using PennyLane's `default.qubit` state-vector simulator (v0.45.1). Kernel evaluation is batched: PennyLane's broadcasting support allows many $(\mathbf{x}_1, \mathbf{x}_2)$ pairs to be evaluated in a single simulator call and for the symmetric train–train kernel matrix only the upper triangle is computed and mirrored, roughly halving the required circuit evaluations. On the 480-sample training split used in this study, the full 480×480 train–train and 120×480 test–train kernel matrices were computed in approximately 15 s on a single CPU core.

4.5 | Classification

The precomputed kernel matrix is passed to a classical Support Vector Machine (scikit-learn's `SVC` with `kernel = "precomputed"`, box constraint $C = 1.0$, Platt-scaled probability outputs), which learns a maximum-margin decision boundary over the quantum-kernel-induced feature space. Because the SVM's convex optimisation problem is solved classically and no circuit parameters are trained, no variational training loop is involved in this QSVM formulation. Section 7 discusses what can and cannot be concluded from the comparison with the VQC.

4.6 | Baselines

Three baselines are trained on the identical PCA-reduced, scaled 8-dimensional feature space for a controlled comparison:

- *Classical Logistic Regression* (L2-regularised, scikit-learn defaults, `max_iter = 1000`).
- *Classical RBF-kernel SVM* (scikit-learn `SVC` with `kernel = "rbf"`), representing the closest classical analogue to the QSVM—both are kernel machines, differing only in the kernel function.
- *Variational Quantum Classifier (VQC)* with the same 8-qubit AngleEmbedding input encoding followed by a 3-layer StronglyEntanglingLayers ansatz, a Pauli-Z expectation readout and a sigmoid output trained with Adam (binary cross-entropy loss, 60 epochs, batch size 16) via PennyLane's PyTorch interface—comparing a trainable circuit with a fixed quantum kernel under otherwise identical preprocessing. Only this single VQC architecture was evaluated.

5 | Experimental Setup

All experiments use the dataset described in Section 3. Records are split 80%/20% (480 train/120 test) using stratified sampling to preserve the 50/50 class balance in both splits (`random_state = 42` throughout for reproducibility). In addition to the single held-out split, we report 5-fold stratified cross-validation across the full dataset for every model (QSVM, Logistic Regression, RBF-SVM and VQC), with a fixed random seed (42), to reduce sensitivity to any one split; the reported spread is the population standard deviation over the five folds. BanglaBERT embeddings are computed once for the full corpus and reused across the held-out split and every cross-validation fold, since the pretrained encoder itself is not fine-tuned or fold-specific; PCA and the MinMax scaler (and, in the TF-IDF ablation, the TF-IDF vectorizer) are refit independently within each split and each fold to prevent information leakage from the test/validation portion. All models were run on a single CPU core; the full pipeline (embedding extraction, kernel construction, training and 5-fold cross-validation) completes in approximately 8–10 min.

Evaluation metrics are accuracy, precision, recall, F1-score and ROC-AUC, computed with the AGT class (Label = 1) as the positive class, alongside the full confusion matrix. Per-genre accuracy is computed from the pooled out-of-fold predictions of the 5-fold protocol, so that every text is predicted exactly once by a model that has not seen it and is reported with 95% Wilson score intervals [34]. Implementation used PennyLane 0.45.1 [35], PyTorch, scikit-learn and the Hugging Face `transformers` library (BanglaBERT loaded from local weights); the additional experiments added in this revision (classical-baseline cross-validation, the TF-IDF ablation and the per-genre analysis) evaluate the kernel of Equations (1) and (2) with an exact NumPy state-vector simulation, which is mathematically equivalent to `default.qubit` for this noiseless circuit. Code and logs are released alongside this paper.

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6 | Results and Analysis

6.1 | Held-Out Test Split

Table 2 reports performance on the single 120-sample held-out test split, comparing the proposed QSVM against classical Logistic Regression, classical RBF-SVM and the VQC baseline, all trained on identical BanglaBERT → PCA(8) → [0,π] features.

The proposed QSVM reaches 95.0% accuracy and 0.989 ROC-AUC, the highest ROC-AUC of the four models, ahead of classical RBF-SVM (0.987) by a margin (0.002) too small to be meaningful. On this 120-sample split, the QSVM's accuracy corresponds to 114 correct predictions, against 115 for Logistic Regression, 112 for RBF-SVM and 105 for the VQC; differences of one or two test samples among the QSVM, Logistic Regression and RBF-SVM are within sampling noise and are not interpreted as a ranking. Figure 2 shows the standard confusion matrices of the QSVM (57 true negatives, 3 false positives, 3 false negatives, 57 true positives) and of the VQC (53, 7, 8, 52). The VQC reaches 87.5% accuracy; its false-positive and false-negative counts (7 and 8) are close, so its errors are roughly symmetric and we draw no conclusion about asymmetry. No calibration-specific evaluation (calibration curve, Brier score, or expected calibration error) was performed, so we make no claim about the calibration of either model. ROC curves are shown in Figure 3 and a summary comparison of the four models in Figure 4.

6.2 | 5-Fold Stratified Cross-Validation

To assess robustness beyond a single split, Table 3 reports 5-fold stratified cross-validation accuracy for all four models under the same protocol; the Logistic Regression and RBF-SVM rows were added in this revision, replacing the earlier single-split reference values.

The QSVM's cross-validated mean accuracy (95.2% ± 2.8%) is consistent with its single-split result (95.0%), suggesting the reported performance is not an artefact of a favourable split. Under the same protocol, Logistic Regression reaches 95.0% ± 2.0% and RBF-SVM 93.8% ± 2.4%, while the VQC reaches 85.8% ± 4.1%. Each validation fold contains 120 texts, so a single text corresponds to 0.83 points and the fold-to-fold standard deviations of the QSVM and classical models are about two to three points; with only five folds we do not attempt a formal significance test and differences among the QSVM, Logistic Regression and RBF-SVM that are small relative to this spread are not interpreted as a ranking. The QSVM was more accurate than the VQC in each of the five folds (by 7.5 to 12.5 points) and its worst fold (90.8%, Fold 4) is above the VQC's mean.

TABLE 2 | Held-out test set performance (BanglaBERT features, 8 qubits/8 PCA components).

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.958	0.951	0.967	0.959	0.972
Classical SVM (RBF)	0.933	0.948	0.917	0.932	0.987
VQC (trainable circuit)	0.875	0.881	0.867	0.874	0.943
QSVM (proposed)	**0.950**	**0.950**	**0.950**	**0.950**	**0.989**

Note: The bold text represents the proposed model's performances.

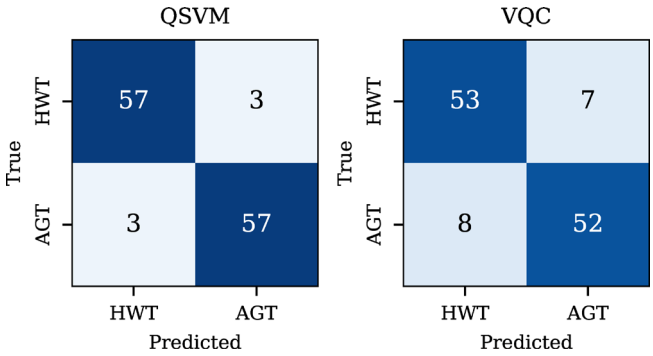


FIGURE 2 | Confusion matrices on the 120-sample held-out split (rows: True class, columns: Predicted class; HWT = human-written, AGT = AI-generated) for the QSVM (left) and the VQC (right).

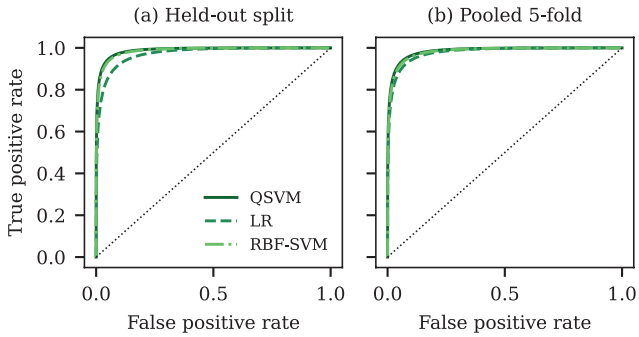


FIGURE 3 | ROC curves for the QSVM, Logistic Regression (LR) and RBF-SVM on identical BanglaBERT → PCA(8) features: (a) held-out split; (b) pooled out-of-fold predictions of the 5-fold protocol. The VQC is omitted because its per-sample scores were not retained; its held-out ROC-AUC is 0.943 (Table 2).

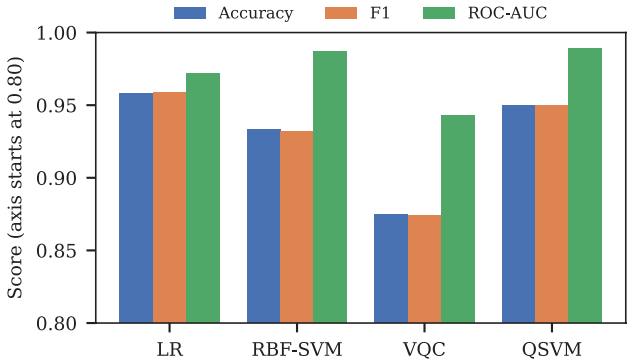


FIGURE 4 | Held-out accuracy, F1-score and ROC-AUC of the four models on identical features (values from Table 2; the vertical axis starts at 0.80).

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6.3 | Effect of the Feature Extractor: TF-IDF vs. BanglaBERT

To examine the role of the feature extractor, we repeated the pipeline (PCA → quantum feature map → classifier) with sparse TF-IDF features in place of BanglaBERT embeddings, holding the qubit count, circuit architecture and training procedure fixed. The ablation covers the QSVM, which is the primary method of this paper, as well as the VQC, with Logistic Regression and RBF-SVM as classical references (Table 4).

For the QSVM, replacing BanglaBERT with TF-IDF reduced held-out accuracy from 95.0% to 89.2% and AGT recall from 95.0% to 83.3%, but changed ROC-AUC only slightly (0.989 to 0.984); its 5-fold cross-validated accuracy fell from 95.2% ± 2.8% to 93.7% ± 2.7%, a difference of about 1.5 points that is smaller than the fold-to-fold standard deviation. For the VQC, single-split accuracy fell from 87.5% to 75.8% and AGT recall from 86.7% to 53.3%, with the resulting confusion matrix skewing heavily towards predicting the HWT class. On the held-out split, BanglaBERT features gave higher accuracy, AGT recall and ROC-AUC than TF-IDF features for every model in Table 4, but the size of the benefit depends on the model and on the evaluation protocol: it is largest for the VQC and for single-split accuracy and small for the QSVM's cross-validated accuracy and ROC-AUC.

Eight principal components retain 74.4% of the variance of the BanglaBERT embeddings but only 13.0% of the TF-IDF variance. This is consistent with the better BanglaBERT results, but our experiments do not establish that retained variance is the cause of the performance difference: the two representations also differ in what they encode (contextual versus lexical information) and these factors are confounded. The TF-IDF VQC result is taken from the original experiment, whose vectorizer configuration retained 13.2% of the variance against 13.0% for the configuration used for the other rows of Table 4; we regard the two as very similar but not identical.

6.4 | Per-Genre Results

Because aggregate accuracy is dominated by newspaper articles and scientific abstracts (80% of the texts), Table 5 breaks the pooled out-of-fold predictions of the 5-fold protocol down by genre. Each entry rests on the n texts of that genre and the Wilson intervals show how much the smaller genres can support: for office orders ($n = 6$, three per class) a single error moves accuracy by 17 points and the interval spans most of the unit range and poems and short stories ($n = 30$ each) are only slightly more precise. Statements about robustness across genres are therefore limited to what these intervals support and we do not claim uniform performance across genres. For the QSVM, pooled out-of-fold accuracy is highest for the two largest genres (97.5% for newspaper articles and 98.1% for scientific abstracts) and lowest for poems (80.0%) and emails (85.2%), the genres with the shortest texts (median lengths of 166 and 197 characters) and Logistic Regression and RBF-SVM show the same ordering; the intervals for poems, short stories and office orders each span more than 20 points.

TABLE 3 | 5-fold stratified cross-validation accuracy (all models, same protocol; SD is the population SD over folds).

Model	F1	F2	F3	F4	F5	Mean ± SD
QSVM	**0.942**	**0.967**	**0.992**	**0.908**	**0.950**	**0.952 ± 0.028**
Logistic Regression	0.942	0.967	0.975	0.917	0.950	0.950 ± 0.020
Classical SVM (RBF)	0.925	0.958	0.967	0.900	0.942	0.938 ± 0.024
VQC	0.867	0.867	0.908	0.783	0.867	0.858 ± 0.041

Note: The bold text represents the proposed model's performances.

TABLE 4 | Feature-extractor ablation: BanglaBERT vs. TF-IDF (8 qubits/8 PCA components; held-out split and 5-fold cross-validation; VQC evaluated on the single split only for TF-IDF).

Model	Features	Acc.	AGT Rec.	AUC	5-fold CV Acc.
QSVM	BanglaBERT	0.950	0.950	0.989	0.952 ± 0.028
QSVM	TF-IDF	0.892	0.833	0.984	0.937 ± 0.027
VQC	BanglaBERT	0.875	0.867	0.943	0.858 ± 0.041
VQC	TF-IDF	0.758	0.533	n/a	n/a
Logistic regression	BanglaBERT	0.958	0.967	0.972	0.950 ± 0.020
Logistic regression	TF-IDF	0.875	0.817	0.951	0.910 ± 0.029
Classical SVM (RBF)	BanglaBERT	0.933	0.917	0.987	0.938 ± 0.024
Classical SVM (RBF)	TF-IDF	0.883	0.817	0.979	0.935 ± 0.032

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TABLE 5 | Per-genre accuracy, % (95% Wilson interval), from pooled out-of-fold predictions (BanglaBERT features).

Genre	*n*	QSVM	LR	RBF-SVM
Newspaper article	318	97.5 (95–99)	97.2 (95–99)	96.2 (94–98)
Scientific abstract	162	98.1 (95–99)	98.1 (95–99)	97.5 (94–99)
Email	54	85.2 (73–92)	83.3 (71–91)	81.5 (69–90)
Poem	30	80.0 (63–90)	83.3 (66–93)	80.0 (63–90)
Short story	30	90.0 (74–97)	90.0 (74–97)	86.7 (70–95)
Office order	6	83.3 (44–97)	83.3 (44–97)	83.3 (44–97)

TABLE 6 | Comparison with classical and transformer baselines on the same dataset (macro-averaged metrics).

Model	Features	Accuracy	Precision	Recall	F1-score
Multinomial Naive Bayes [11]	TF-IDF	0.742	0.869	0.521	0.465
Random Forest [11]	TF-IDF	0.753	0.874	0.542	0.505
SVM (polynomial, degree 6) [11]	TF-IDF	0.764	0.382	0.500	0.433
KNN ($k = 15$) [11]	TF-IDF	0.809	0.737	0.694	0.710
CNN–LSTM [11]	BanglaBERT embeddings	0.764	0.382	0.500	0.433
Fine-tuned BanglaBERT [11]	BanglaBERT (fine-tuned)	**0.955**	**0.972**	0.905	0.933
QSVM (proposed)	BanglaBERT + PCA(8) + quantum kernel	0.950	0.950	**0.950**	**0.950**

Note: The bold text represents the proposed model's performances.

6.5 | Comparison With Existing Literature on the Same Dataset

To situate the proposed QSVM relative to a wider range of non-quantum approaches, Table 6 compares it against six classifiers—Multinomial Naive Bayes, Random Forest, SVM, KNN, a CNN–LSTM head and a fine-tuned BanglaBERT model—reported for the same dataset in the paper that originally introduced it [11], using TF-IDF features (word- and character-level *n*-grams) for the classical models and BanglaBERT-derived representations for the deep learning models. These baselines were trained under a different protocol from the QSVM and its in-pipeline comparators of Section 6.1—a 70%/30% train–test split (stratified for the SVM and KNN models only) rather than the 80%/20% stratified split used for the QSVM and no PCA-based qubit-count restriction—so the comparison should be read as indicative of relative model families rather than as a fully controlled ablation. In particular, in our own controlled runs (Table 6) TF-IDF features compressed to 8 principal components already support 87.5%–89.2% held-out accuracy for the QSVM, Logistic Regression and RBF-SVM, well above the 74%–81% of the TF-IDF baselines reported in [11]; the gap between those baselines and the QSVM in Table 6 therefore cannot be attributed to the quantum kernel or to BanglaBERT alone and also reflects differences in split, feature construction and classifier configuration.

The classical TF-IDF baselines cluster in the 74%–81% accuracy range, with KNN the strongest of the four (80.9% accuracy, 0.710 F1); the polynomial-kernel SVM and the CNN–LSTM head over BanglaBERT embeddings both collapse to an identical, degenerate operating point (76.4% accuracy, 0.433 F1, precision far below recall), consistent with a classifier that has converged to predicting a single majority-leaning class rather than learning a genuine decision boundary. The fully fine-tuned BanglaBERT classifier is the strongest model overall (95.5% accuracy, 0.933 F1), marginally ahead of the proposed QSVM (95.0% accuracy, 0.950 F1). This is an expected ordering: full fine-tuning updates all ~110M BanglaBERT parameters directly on the task, whereas the QSVM operates on a frozen, 8-dimensional PCA compression of the same embedding space passed through a fixed, non-trainable quantum feature map. That the QSVM nonetheless comes within one accuracy point of fully fine-tuned BanglaBERT and exceeds the classical TF-IDF baselines and the BanglaBERT-embedding-based CNN–LSTM of [11] under their protocol, suggests that a compact 8-dimensional projection of frozen BanglaBERT embeddings retains most of the discriminative signal that full fine-tuning captures. Because Logistic Regression and RBF-SVM on the same projection perform comparably (Table 2), this is better credited to the frozen features and the projection than to the quantum kernel specifically.

7 | Discussion

Four observations stand out. First, under the specific configurations evaluated, the QSVM (a fixed 8-qubit kernel with $C = 1$) was more accurate than the VQC (a single 3-layer StronglyEntanglingLayers ansatz) under an otherwise identical pipeline—the QSVM's cross-validated accuracy (95.2%) exceeds the VQC's (85.8%) by more than nine points, with

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lower variance across folds and a higher accuracy in every fold. One candidate explanation is that trainable circuits are prone to flat loss landscapes and poor local minima on small datasets [17, 30], whereas kernel methods delegate the (convex, well-understood) optimisation problem to a classical SVM. Our experiments do not test this explanation: we did not analyse gradient magnitudes or loss landscapes, did not vary the ansatz depth, optimiser settings, or number of epochs and did not include a control that separates optimisation from expressivity, so other factors may account for the gap. Because only one VQC architecture was evaluated, the results do not show that fixed quantum kernels are generally superior to trainable variational circuits; they show that this QSVM outperformed this VQC on this task.

Second, the QSVM is competitive with strong classical baselines trained on the same 8-dimensional feature space—held-out ROC-AUC 0.989 against 0.972 for Logistic Regression and 0.987 for RBF-SVM, accuracy 95.0% against 95.8% and 93.3% and the cross-validated accuracies of Table 3—despite operating with an 8-qubit bottleneck that discards over 25% of the BanglaBERT embedding's variance. These differences are small relative to the fold-to-fold variability, so the evidence supports comparability rather than superiority or inferiority of the QSVM with respect to these classical models. The absence of a clear quantum advantage is an expected outcome at present qubit counts: any quantum advantage from kernel methods is theorised to emerge primarily in regimes where the target function is genuinely hard to approximate with a classical kernel [12], which a moderately-separable binary text-classification task under eight features is unlikely to be. The value of this result is therefore less a claim of outright superiority over classical methods and more a demonstration that a well-designed quantum kernel pipeline is a practically viable, competitive alternative, not merely a theoretical curiosity, even on today's classical simulators.

Third, the feature-extractor ablation (Section 6.3) shows that BanglaBERT features gave better held-out accuracy, AGT recall and ROC-AUC than TF-IDF features for every model tested, but that the effect on the QSVM's cross-validated accuracy was modest (about 1.5 points, less than one standard deviation). We therefore do not claim that the feature extractor matters more than choices within the quantum circuit: the feature map, entangling topology and scaling were not varied. What the data support is that BanglaBERT is a stronger starting representation in this setting and that the size of its advantage depends on the model and on the evaluation protocol. Because the number of qubits realistically simulable today caps the number of PCA components that can be encoded, dense contextual embeddings such as BanglaBERT remain a natural choice for the pre-reduction representation.

Fourth, the strong genre imbalance of the dataset (Table 1) means that the aggregate metrics mostly describe newspaper articles and scientific abstracts. The per-genre results (Table 5) are informative for the larger genres but too imprecise for poems, short stories and especially office orders (six texts) to support any claim of robustness across genres; a larger and more balanced genre mix would be needed for that.

8 | Limitations

8.1 | Simulation, Not Quantum Hardware

All quantum circuits were executed on a classical state-vector simulator (PennyLane `default.qubit`); no results here reflect noisy intermediate-scale quantum (NISQ) hardware behaviour, where gate errors, decoherence and measurement noise would likely degrade the fidelity kernel's precision.

8.2 | Qubit-Bounded Compression

The 8-qubit budget forces an aggressive 768-to-8 dimensionality reduction of the BanglaBERT embedding, discarding roughly a quarter of its variance; a higher qubit count (tractable on real quantum hardware, though not efficiently simulable classically at scale) could retain more signal.

8.3 | Single Generator Family

As with the underlying dataset [11], generalisation to AI text from generators not represented in the training distribution is untested, mirroring a limitation broadly reported across the AI-text detection literature [1, 7, 18]. The AI-generated texts come from a single system (ChatGPT) and the human-written texts from a limited set of sources, so source-specific style cues may contribute to detectability.

8.4 | Genre Imbalance and Small Per-Genre Samples

The dataset contains 318 newspaper articles and 162 scientific abstracts but only 54 emails, 30 poems, 30 short stories and six office orders (Table 1). Aggregate metrics are dominated by the two largest genres; per-genre estimates for the small genres have wide confidence intervals (Table 5) and the office-order estimate rests on six texts. Our results therefore do not establish robustness across genres.

8.5 | Single VQC Architecture and Single QSVM Configuration

Only one VQC architecture (3-layer StronglyEntanglingLayers, 60 epochs) and one QSVM configuration (angle-embedding feature map with a CZ ring, $C = 1$, 8 qubits) were evaluated. The QSVM–VQC comparison holds only for these configurations and the cause of the gap between them was not investigated.

8.6 | No Calibration Analysis

We did not evaluate calibration (e.g., calibration curves, Brier score, or expected calibration error) and make no claim about the calibration of any model.

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9 | Conclusion and Future Work

This paper presented, to the best of our knowledge, the first application of a quantum kernel Support Vector Machine (QSVM) to the detection of AI-generated Bangla text. Coupling BanglaBERT contextual embeddings with an 8-qubit angle-embedding quantum feature map and a fidelity-based kernel, the proposed QSVM achieved 95.0% accuracy and 0.989 ROC-AUC on a held-out split and 95.2% (±2.8%) mean accuracy under 5-fold cross-validation—comparable to classical Logistic Regression (95.0% ± 2.0%) and RBF-SVM (93.8% ± 2.4%) evaluated under the same protocol and more accurate than the single Variational Quantum Classifier evaluated (85.8% ± 4.1%). A TF-IDF ablation shows that BanglaBERT features helped every model on the held-out split, with a modest effect on the QSVM's cross-validated accuracy and per-genre results show that the strongly imbalanced genre mix supports conclusions mainly for newspaper articles and scientific abstracts. Within the configurations studied, these results suggest that a fixed quantum kernel is a practical entry point for QML on small, low-resource NLP datasets; they do not show that fixed kernels are generally better than trainable variational circuits, nor that the QSVM outperforms strong classical baselines.

Future work includes evaluating on NISQ hardware to characterise the effect of realistic gate noise on the fidelity kernel; exploring higher qubit counts and alternative entangling topologies (linear, all-to-all) as simulator and hardware capacity permits; extending the dataset with additional genres (with more balanced genre sizes) and a wider range of generator models to test cross-generator generalisation, consistent with recommendations from the broader AI-text detection literature [1, 18, 23]; benchmarking against more encoders; evaluating additional VQC architectures and analysing their optimisation behaviour; and investigating hybrid quantum–classical ensembles that combine the QSVM's kernel with classical model outputs.

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